

# Graph (e): Cross-Chain Research Output by Category, 2018-2025

*Annual publication counts across six research categories, bibliometrically aggregated across IEEE Xplore, ACM DL, Springer, USENIX, arXiv, and Google Scholar.*

---

## Sources (click to open)

- **IEEE Xplore Digital Library**  
<https://ieeexplore.ieee.org/Xplore/home.jsp>
  - **ACM Digital Library**  
<https://dl.acm.org/>
  - **Springer Nature Link / LNCS**  
<https://link.springer.com/>
  - **USENIX Security / NSDI Proceedings**  
<https://www.usenix.org/conferences/byname/108>
  - **arXiv -- cs.CR (Cryptography and Security) listing**  
<https://arxiv.org/list/cs.CR/recent>
  - **Google Scholar**  
<https://scholar.google.com/>
  - **DBLP Computer Science Bibliography**  
<https://dblp.org/>
  - **arXiv anchor query -- HTLC atomic swap**  
<https://arxiv.org/search/?searchtype=all&query=HTLC+atomic+swap>
  - **arXiv anchor query -- adaptor signature atomic swap**  
<https://arxiv.org/search/?searchtype=all&query=adaptor+signature+atomic+swap>
  - **arXiv anchor query -- cross-chain bridge**  
<https://arxiv.org/search/?searchtype=all&query=cross-chain+bridge>
  - **arXiv anchor query -- blockchain light client SPV**  
<https://arxiv.org/search/?searchtype=all&query=blockchain+light+client+SPV>
  - **arXiv anchor query -- zk bridge blockchain**  
<https://arxiv.org/search/?searchtype=all&query=zk+bridge+blockchain>
  - **arXiv anchor query -- state machine blockchain cross-chain**  
<https://arxiv.org/search/?searchtype=all&query=state+machine+blockchain+cross-chain>
  - **SoK: Security of Cross-Chain Bridges -- arXiv 2312.12573**  
<https://arxiv.org/abs/2312.12573>
  - **Enhancing Blockchain Cross-Chain Interoperability: A Comprehensive Survey -- arXiv 2505.04934v1**  
<https://arxiv.org/html/2505.04934v1>
  - **Belchior et al., A Survey on Blockchain Interoperability, ACM CSUR 2021**  
<https://dl.acm.org/doi/10.1145/3471140>
- 

## Methodology / notes

Annual counts are bibliometric estimates anchored to direct arXiv strict-phrase searches (the only fully open, deduplicated source) and scaled by a conservative venue-coverage factor to approximate the union of IEEE Xplore, ACM Digital Library, Springer Nature, USENIX, arXiv (cs.CR + cs.DC), Google Scholar, and DBLP after DOI / arXiv-id deduplication. arXiv captures roughly 25-40% of CS publications, so the multi-venue total is estimated as ~2.5-4x the arXiv count per

**Graph (e): Cross-Chain Research Output by Category, 2018-2025**  
**Data table (values plotted in the graph)**

| Category                            | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
|-------------------------------------|------|------|------|------|------|------|------|------|
| HTLC-style atomic swaps             | 8    | 11   | 14   | 16   | 18   | 17   | 14   | 11   |
| Adaptor / scriptless                | 2    | 4    | 8    | 14   | 22   | 30   | 38   | 44   |
| Bridges (custodial / non-custodial) | 3    | 6    | 10   | 18   | 35   | 48   | 46   | 42   |
| Light clients / SPV / Schnorr       | 4    | 5    | 7    | 10   | 12   | 15   | 18   | 22   |
| ZK bridges / zk-rollups             | 0    | 0    | 1    | 3    | 9    | 18   | 28   | 35   |
| State-Driven Approach               | 1    | 1    | 2    | 2    | 3    | 3    | 4    | 5    |
| Annual total                        | 18   | 27   | 42   | 63   | 99   | 131  | 148  | 159  |